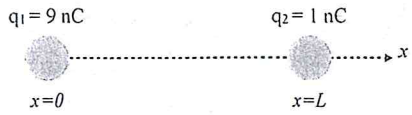
Problem 1 (15 points)

Two point charges with $q_1 = 9$ nC and $q_2 = 1$ nC are pivoted as shown in the figure. Another point particle with an unknown charge should be placed on the x-axis somewhere in between these two charges. Find the x coordinate such that the particle with unknown charge stays at rest.



Net force acting on "q" must be zero if it stays at rest.

$$\sum \vec{F} = \frac{1}{4\pi\epsilon_0}\left[\frac{qq_1}{x^2} - \frac{qq_2}{(x-L)^2}\right] = 0$$

from the equation above

$$\frac{9}{x^2} - \frac{1}{(x-L)^2} = 0 \Rightarrow \frac{9}{x^2} = \frac{1}{(x-L)^2}$$

$$\Rightarrow 9(x-L)^2 = x^2$$

$$\Rightarrow 3(x-L) = x \quad \text{or} \quad 3(x-L) = -x$$

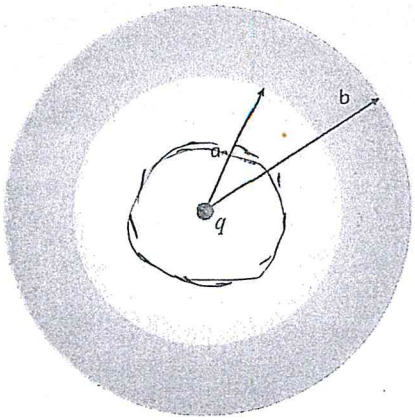
$$\Rightarrow 3x - 3L = x \qquad 3x - 3L = -x$$

$$\Rightarrow 2x = 3L \qquad 4x = 3L$$

$$\Rightarrow \underbrace{x = \frac{3L}{2}}_{\text{it is not in between } q_1 \text{ and } q_2} \qquad \boxed{x = \frac{3L}{4}}$$

it is in between

Problem 2 (30 points)



A neutral conducting spherical shell with inner radius a and outer radius b encloses a point particle with positive charge q placed right at the center of the spherical shell. Find the electric field as a function of the radial distance "r" from the center of the sphere for the following regions:

(a) (5 pts.) $r < a$,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$ take a Gaussian surface with $r < a$

$$E(r)\, 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow \boxed{\vec{E}(r) = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{r}}$$

(b) (5 pts.) $a < r < b$,

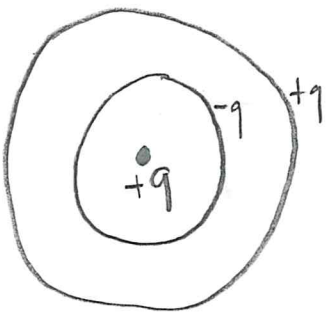
$\boxed{\vec{E}(r) = 0}$ inside the conducting material.

(c) (10 pts.) $b < r$.

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$ take a Gaussian surface with $r > b$

$$\boxed{\vec{E}(r) = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{r}}$$

(d) (10 pts.) Find the inner and outer surface charges of the shell.

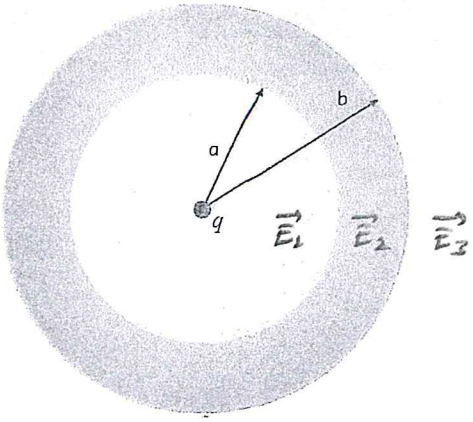


Charge on the inner surface : $\boxed{-q}$

Charge on the outer surface : $\boxed{+q}$

Problem 3 (25 points)

A neutral conducting spherical shell with inner radius a and outer radius b encloses a point particle with positive charge q placed right at the center of the spherical shell. Find the electric potential as a function of the radial distance "r" from the center of the sphere for the following regions:



(a) (10 pts.) $r < a$,

$$V_r - \cancel{V_\infty}^{\,0} = \int_r^a \vec{E}_1 \cdot d\vec{l} + \cancel{\int_a^b \vec{E}_2 \cdot d\vec{l}}^{\,0} + \int_b^\infty \vec{E}_3 \cdot d\vec{l}$$

$$V_r = \frac{q}{4\pi\epsilon_0}\left[\int_r^a \frac{dr}{r^2} + \int_b^\infty \frac{dr}{r^2}\right] = \frac{q}{4\pi\epsilon_0}\left[\left(-\frac{1}{r}\right)\Big|_r^a + \left(-\frac{1}{r}\right)\Big|_b^\infty\right]$$

$$= \frac{q}{4\pi\epsilon_0}\left[-\frac{1}{a} + \frac{1}{b} + \frac{1}{r}\right]$$

(b) (10 pts.) $a < r < b$,

$$V_r - \cancel{V_\infty}^{\,0} = \cancel{\int_r^b \vec{E}_2 \cdot d\vec{l}}^{\,0} + \int_b^\infty \vec{E}_3 \cdot d\vec{l}$$

$$V_r = \frac{q}{4\pi\epsilon_0}\int_b^\infty \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0}\left(-\frac{1}{r}\Big|_b^\infty\right) = \boxed{\frac{q}{4\pi\epsilon_0}\cdot\frac{1}{b}}$$

(c) (5 pts.) $b < r$.

$$V_r - \cancel{V_\infty}^{\,0} = \int_r^\infty \vec{E}_3 \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0}\int_r^\infty \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0}\left(-\frac{1}{r}\Big|_r^\infty\right)$$

$$V_r = \frac{q}{4\pi\epsilon_0}\left(\cancel{-\frac{1}{\infty}}^{\,0} + \frac{1}{r}\right)$$

$$\boxed{V_r = \frac{q}{4\pi\epsilon_0}\cdot\frac{1}{r}}$$

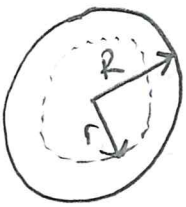
Problem 4 (30 points)

An insulating sphere of radius R has non-uniform charge distributed over its volume. The charge density is given as $\rho = \alpha r$, where r is the radial distance from the center of the sphere.

(a) (5 pts.) What is the unit of α in SI unit system?

$\rho = \frac{Q}{V} \left[\frac{C}{m^3}\right]$ $\quad \left[\frac{C}{m^3}\right] = [?][m] \Rightarrow [?] = \left[\frac{C}{m^4}\right]$

(b) (10 pts.) Find the electric field for $r < R$.



$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$

$E(r) 4\pi r^2 = \frac{q_{enc}}{\epsilon_0}$

$E(r) 4\pi r^2 = \frac{\pi \alpha r^4}{\epsilon_0}$

$\boxed{E(r) = \frac{\alpha}{4\epsilon_0} r^2 \hat{r}}$

$\rho = \frac{dq}{dV} \Rightarrow dq = \int \rho \, dV$

$q_{enc} = \int \alpha r \, dV$

$V = \frac{4}{3}\pi r^3 \Rightarrow dV = 4\pi r^2$

$q_{enc} = \int_0^r \alpha r 4\pi r^2 dr$

$q_{enc} = \pi \alpha r^4$

(c) (15 pts.) Find the work that must be done by an external force to move a proton with charge q_p, from the surface of the sphere ($r = R$), to a point inside the sphere at $r = R/2$.

$W_{\text{Electric Field}} + W_{\text{External Force}} = 0 \Rightarrow W_{\text{External force}} = -W_{\text{Electric Field}}$

$W_{\text{Electric Field}} = -\Delta U_E = -q_p \Delta V = -q_p (V_f - V_i)$

$W_{\text{Elec. Field}} = -q_p \int_f^i \vec{E} \cdot d\vec{l}$

$W_{\text{Elec. Field}} = -q_p \int_{R/2}^{R} \left(\frac{\alpha}{4\epsilon_0} r^2 \hat{r}\right) \cdot (dr \, \hat{r})$

$W_{\text{Elec. Field}} = -q_p \int_{R/2}^{R} \frac{\alpha}{4\epsilon_0} r^2 dr$

$W_{\text{Electric Field}} = -\frac{7\alpha q_p R^3}{96\epsilon_0}$

$\boxed{W_{\text{External Force}} = \frac{7\alpha q_p R^3}{96\epsilon_0}}$